

RHIE — Rail Health Intelligence Engine

Complete Technical Dossier

Competition: Far Away 2026, organised by Zuup

Team: Moshi Moshi Japan

Project: Physics-Informed Predictive Maintenance Dashboard for Indian Railways Track

Rail Standard: GR1080 IRS T12-2009 (Nippon Steel K003en verified)

Coverage: 20 IMD Climate Zones · 68 IR Divisions · 7 Route Classes · 48 IRPWM Repair Methods

Implementation: React + Recharts (browser dashboard) + scikit-learn GradientBoosting (Python validation)

Table of Contents

1. [Project Overview](#)
2. [Material Specification — GR1080 Rail](#)
3. [Rail Section Properties](#)
4. [Route Classification and Traffic Loading](#)
5. [GMT Life Limits and Wear Limits](#)
6. [Physics Engine](#)
 - [6.1 Hertz Contact Mechanics](#)
 - [6.2 Winkler Foundation Model \(Rail Bending\)](#)
 - [6.3 CWR Thermal Stress Analysis](#)
 - [6.4 Paris Law Fatigue Crack Propagation](#)
 - [6.5 GMT Wear Model](#)
7. [Indian Climate Zones — All 20](#)
8. [Indian Railways Zones and Division–Climate Mapping](#)
9. [Seasonal Calendar](#)
10. [Health Index — Formula and Scoring](#)
11. [Rail Stage Classification](#)
12. [ML Model — Complete Specification](#)
 - [12.1 Architecture and Training](#)
 - [12.2 Feature Engineering](#)
 - [12.3 Target Variables](#)
 - [12.4 Training Data Generation](#)

- [12.5 Hyperparameters](#)
- [12.6 Training Curve](#)
- [12.7 Regression Metrics — RUL Prediction](#)
- [12.8 Classification Metrics — Defect Type](#)
- [12.9 Per-Class Performance](#)
- [12.10 Confusion Matrix](#)
- [12.11 Feature Importance](#)
- [12.12 What Each Metric Means](#)

13. [Defect Detection and Repair Ranking](#)

14. [All 48 IRPWM Repair Methods](#)

15. [AI Assessment Engine](#)

16. [Dashboard Architecture](#)

1. Project Overview

RHIE is a physics-informed predictive maintenance dashboard for Indian Railways Permanent Way engineers. It runs entirely in the browser — no backend, no database — combining four independent analytical layers:

Layer 1 — Physics Engine: Hertz contact mechanics, Winkler elastic foundation bending, Paris Law fatigue crack propagation, GMT-based wear modelling, and CWR thermal stress analysis. All derived from first principles using material constants verified against IRS T12-2009 and Nippon Steel K003en test data.

Layer 2 — Climate Intelligence: 20 IMD (1991–2020) climate zones covering all major geographic regions of India. Each zone provides monthly ambient temperature, solar gain, annual rainfall, terrain type, seismic classification, and Stress-Free Temperature (SFT) for CWR distressing computations.

Layer 3 — ML Model: A Gradient Boosting model trained on 500 physics-derived synthetic samples (browser) / 1500 balanced samples (Python scikit-learn validation). Two tasks: regression to predict RUL in MGT remaining, and multi-class classification across 5 defect types. The ML runs silently in background on mount and feeds the Defect & Repair tab.

Layer 4 — Rule-Based Assessment: A deterministic IRPWM-grounded assessment engine that synthesises all computed physics data into a structured P-Way report with condition summary, risk ranking, immediate actions, maintenance programme, TSR recommendation, and inspection schedule.

2. Material Specification — GR1080 Rail

All physics computations use GR1080 high-strength pearlitic rail steel per IRS T12-2009. Constants verified against Nippon Steel K003en track test data and University of Wollongong rail fatigue studies.

Property	Symbol	Value	Unit	Source
Young's Modulus	E	210	GPa	IRS T12-2009 §4.3
Poisson's Ratio	ν	0.30	—	IRS T12-2009 §4.3
Thermal Expansion Coefficient	α	11.5×10^{-6}	/°C	IRS T12-2009 §4.4
Yield Strength	F _y	460	MPa	IRS T12-2009 §3.2
Ultimate Tensile Strength	UTS	1080	MPa	GR1080 designation
Hardness (Brinell)	HBW	340–390	—	IRS T12-2009 §3.3
Fracture Toughness	K _{IC}	29	MPa√m	Croft et al. (2016), Aglan (2004)
Paris Law Coefficient	C	1.65×10^{-11}	m/cycle per (MPa√m) ³	Zerbst et al. (2005)
Paris Law Exponent	m	3.0	—	Standard value for rail steels
Elastic Shakedown Limit	p _{0_max}	1062	MPa	Bower-Johnson criterion, GR1080
Tangent Wear Rate	w _t	0.013	mm/MGT	Nippon Steel K003en (DHH370, R=290m baseline)
Mild Curve Wear Rate	w _m	0.025	mm/MGT	Nippon Steel K003en, curve wear tests
Sharp Curve Wear Rate	w _s	0.044	mm/MGT	Nippon Steel K003en (R<400m)

Notes on material constants:

- **K_{IC} = 29 MPa√m** is a conservative value for GR1080 at rail head temperature. Values reported in literature range from 22–35 MPa√m depending on temperature and crack orientation. 29 MPa√m is used for Mode I (opening) transverse cracks at ambient temperature.

- **Paris C** = 1.65×10^{-11} with units m/cycle per $(\text{MPa}\sqrt{\text{m}})^3$ applies specifically for $da/dN = C(\Delta K)^m$ with K in $\text{MPa}\sqrt{\text{m}}$ and crack depth a in metres. Verified against Zerbst et al. (2005) crack growth data for pearlitic rail steel.
- **Wear rates** from Nippon Steel K003en test track (Shinkansen and JR freight tests, 2018) adjusted for BG Indian conditions (heavier axle loads) using RDSO field data.
- **Shakedown limit 1062 MPa** derived from Bower-Johnson criterion: $p_0_{\text{shakedown}} = (3/2) \cdot k_{\text{shakedown}}$ where k is the shear yield stress. For GR1080 with $F_y = 460 \text{ MPa}$: $k = 460/\sqrt{3} = 265.6 \text{ MPa} \rightarrow p_0 = 1.5 \times 1.5 \times k = 1062 \text{ MPa}$ (approximate). When p_0 exceeds this, every wheel pass causes a small increment of plastic deformation at the rail head — ratcheting wear accumulates faster than Paris Law predicts.

3. Rail Section Properties

Two section types are modelled — 60 kg/m (UIC60, most modern Indian main lines) and 52 kg/m (UIC54, older secondary lines).

Property	Symbol	60 kg/m	52 kg/m	Unit
Second Moment of Area	I	3055×10^{-8}	2346×10^{-8}	m^4
Dist. neutral axis → foot (bottom)	y_b	80.0	77.3	mm
Dist. neutral axis → head (top)	y_t	91.6	87.5	mm
Section Height	h	~171.6	~164.8	mm

These properties enter the Winkler bending calculation to compute rail foot bending stress ($\sigma_b = M \cdot y_b / I$) and head compression stress ($\sigma_t = M \cdot y_t / I$) under wheel loading.

4. Route Classification and Traffic Loading

Seven route classes are modelled covering the full spectrum of IR traffic density. Each class specifies freight trains/day, passenger trains/day, EMU trains/day, and the gross tonnage per consist.

Class	Label	Freight (fr)	Passenger (pa)	EMU (em)	Freight Gross (fg, t)	Pax Gross (pg, t)	EMU Gross (eg, t)
HM	Heavy Mineral	80	10	0	3200	1500	0
A+	Group A+ (>25 MGT/yr)	40	60	0	2200	1461	0
A	Group A	20	40	0	2000	1461	0

Class	Label	Freight (fr)	Passenger (pa)	EMU (em)	Freight Gross (fg, t)	Pax Gross (pg, t)	EMU Gross (eg, t)
	(10–25 MGT/yr)						
B	Group B (3–10 MGT/yr)	10	20	0	1800	1300	0
C	Group C (0.5–3 MGT/yr)	3	10	0	1500	1200	0
D	Group D (<0.5 MGT/yr)	1	3	0	1200	1000	0
SUB	Dense Suburban (EMU)	0	20	200	0	1200	540

Daily GMT formula:

$\text{GMT_daily (MGT/day)} = (\text{fr} \times \text{fg} + \text{pa} \times \text{pg} + \text{em} \times \text{eg}) / 1,000,000$
 $\text{GMT_annual (MGT/yr)} = \text{GMT_daily} \times 365$
 $\text{GMT_cumulative (MGT)} = \text{GMT_annual} \times \text{age_in_years}$

Axle pass frequency for Paris Law:

$\text{estFreq (axle passes/day)} = \text{fr} \times 86 + \text{pa} \times 96 + \text{em} \times 48$

Freight consists assumed 86 axles (43 wagons $\times$ 2 bogies $\times$ 1 axle each pair), passenger 96 axles (24 coaches $\times$ 4 axles), EMU 48 axles. This determines how many fatigue cycles the crack experiences per day.

5. GMT Life Limits and Wear Limits

GMT Life Limits (IRPWM Para 302)

Rail Type	No Grinding	With Grinding	Broad Gauge (BG) Only
60 kg/m	800 MGT	1000 MGT	Yes
52 kg/m	525 MGT	700 MGT	Yes

Grinding extends life by delaying the onset of detectable RCF (Rolling Contact Fatigue) by removing the fatigued surface layer before cracks initiate. The dashboard uses the appropriate limit based on user selection of the grinding programme toggle.

Wear Limits (IRPWM, vertical head wear)

Rail Type	Renewal Limit
60 kg/m	8 mm vertical head wear
52 kg/m	13 mm vertical head wear

The higher limit for 52 kg/m reflects the larger head depth available before the neutral axis is approached.

6. Physics Engine

6.1 Hertz Contact Mechanics

Purpose: Compute the peak contact pressure (p_0) at the wheel–rail interface. If p_0 exceeds the elastic shakedown limit (1062 MPa), plastic deformation occurs with every wheel pass — ratcheting regime.

Reference: Hertz (1882), adapted for wheel-rail by Johnson (1985), calibrated to BG Indian Railways (22.9t standard axle load → $a=9\text{mm}$, $b=6\text{mm}$ semi-axes).

Derivation:

A wheel in rolling contact with a rail creates an elliptical contact patch. For the standard BG axle load of 22.9 tonnes:

```
Wheel load per rail: P = axle_load × 1000 × 9.81 / 2    [N]

Contact patch scaling from reference:
  s = (axle_load / 22.9)^(1/3)    (geometric scaling)
  a = 9 mm × s                    (semi-axis along rolling direction)
  b = 6 mm × s                    (semi-axis transverse)

Peak Hertz pressure:
  p_0 = 3P / (2π × a × b)          [Pa → divide by 10^6 for MPa]

Peak shear stress (sub-surface):
  τ_max = 0.31 × p_0              (at depth ≈ 0.48b below surface)

Contact area:
  A = π × a × b                  [mm^2]

Plastic regime flag:
  plastic = (p_0 > 1062 MPa)
```

Example — standard 22.9t BG axle:

- $P = 22.9 \times 1000 \times 9.81 / 2 = 112,319 \text{ N}$
- $s = 1.0 \rightarrow a = 9 \text{ mm}, b = 6 \text{ mm}$
- $p_0 = 3 \times 112,319 / (2\pi \times 0.009 \times 0.006) = \sim 993 \text{ MPa}$
- $\tau_{\text{max}} = 0.31 \times 993 = 308 \text{ MPa}$

- Below shakedown limit (1062 MPa) — elastic regime

Example — heavy freight axle, 30t:

- $P = 30 \times 1000 \times 9.81 / 2 = 147,150 \text{ N}$
- $s = (30/22.9)^{(1/3)} = 1.096 \rightarrow a = 9.87 \text{ mm}, b = 6.58 \text{ mm}$
- $p_0 = 3 \times 147,150 / (2\pi \times 0.00987 \times 0.00658) = \sim 1082 \text{ MPa}$
- **PLASTIC REGIME** — ratcheting wear begins

6.2 Winkler Foundation Model (Rail Bending)

Purpose: Compute the bending moment and resulting stresses in the rail foot and head under wheel loading, accounting for track quality (foundation stiffness), dynamic amplification, and rail geometry.

Reference: Winkler (1867) elastic foundation, Timoshenko (1940) rail bending on elastic support, adapted for IRPWM BG track.

Track Quality Parameters (foundation modulus u):

Track Quality	$u \text{ (N/m}^2\text{)}$	Typical Condition
Excellent	60×10^6	New track, recent deep tamping, premium ballast
Good	45×10^6	Well-maintained, regular tamping, clean ballast
Fair	30×10^6	Moderate maintenance, some ballast fouling
Poor	18×10^6	Significant fouling, deferred maintenance, soft subgrade

Derivation:

Characteristic length (inverse decay length):

$$\lambda = (u / (4 \times E \times I))^{(1/4)} \quad [\text{m}^{-1}]$$

Dynamic Amplification Factor:

$$\text{DAF} = 1 + 0.003 \times V_{\text{max}} \quad (V_{\text{max}} \text{ in km/h})$$

Dynamically amplified wheel load:

$$P_{\text{dyn}} = (\text{axle} \times 1000 \times 9.81 / 2) \times \text{DAF} \quad [\text{N}]$$

Maximum bending moment:

$$M = P_{\text{dyn}} / (4 \times \lambda) \quad [\text{N}\cdot\text{m}]$$

Rail foot tensile stress (governs fatigue):

$$\sigma_b = M \times y_b / I \quad [\text{Pa} \rightarrow \text{MPa}]$$

Rail head compressive stress:

$$\sigma_t = M \times y_t / I \quad [\text{Pa} \rightarrow \text{MPa}]$$

Effective stress range for Paris Law:

$$\Delta\sigma = (M \times y_b / I + 0.3 \times M \times y_t / I) / 10^6 \quad [\text{MPa}]$$

The 0.3 factor on the head compressive stress represents the contribution of rail head compression to the equivalent damage-causing stress range at the foot. This captures the combined tension–compression cycle experienced at the rail foot under a passing wheel.

Example — Route A, 60 kg/m, fair quality, V=110 km/h, 22.9t axle:

- $\lambda = (30 \times 10^6 / (4 \times 210 \times 10^9 \times 3055 \times 10^{-8}))^{0.25} = 1.040 \text{ m}^{-1}$
- $\text{DAF} = 1 + 0.003 \times 110 = 1.33$
- $P_{\text{dyn}} = 22.9 \times 1000 \times 9.81 / 2 \times 1.33 = 149,463 \text{ N}$
- $M = 149,463 / (4 \times 1.040) = 35,929 \text{ N}\cdot\text{m}$
- $\sigma_b = 35,929 \times 0.080 / 3055 \times 10^{-8} = 94.1 \text{ MPa}$
- $\sigma_t = 35,929 \times 0.0916 / 3055 \times 10^{-8} = 107.7 \text{ MPa}$
- $\Delta\sigma = (94.1 + 0.3 \times 107.7) = 126.4 \text{ MPa}$

Thermal stress is then added to $\Delta\sigma$ for the Paris Law calculation (see §6.4), but NOT for the health score crack fraction (which uses bending-only $\Delta\sigma$ to avoid seasonal health oscillation).

6.3 CWR Thermal Stress Analysis

Purpose: Compute the thermal axial stress in Continuously Welded Rail (CWR) due to temperature deviation from the Stress-Free Temperature (SFT). Excessive compressive stress (hot rail) causes CWR buckling; excessive tensile stress (cold rail) causes transverse fracture.

Reference: IRPWM §7, IR Code for CWR, Esvelde (2001) Modern Railway Track.

Derivation:

CWR is welded at the SFT — the temperature at which the rail is stress-free. Any deviation from SFT causes locked-in thermal stress:

Rail temperature:

$$T_{\text{rail}} = T_{\text{ambient}} + T_{\text{solar_gain}} \quad [^{\circ}\text{C}]$$

(from IMD monthly climate data per zone)

Temperature deviation from SFT:

$$\Delta T = T_{\text{rail}} - \text{SFT} \quad [^{\circ}\text{C}]$$

Thermal stress:

$$\sigma_{\text{th}} = E \times \alpha \times \Delta T \quad [\text{Pa} \rightarrow \text{MPa}]$$

$$\sigma_{\text{th}} = 210,000 \text{ MPa} \times 11.5 \times 10^{-6} / ^{\circ}\text{C} \times \Delta T$$

Risk classification:

$$\begin{aligned} |\Delta T| < 8^{\circ}\text{C} &\rightarrow \text{LOW} && (\text{normal operating range}) \\ |\Delta T| < 15^{\circ}\text{C} &\rightarrow \text{MODERATE} && (\text{increased monitoring}) \\ |\Delta T| < 23^{\circ}\text{C} &\rightarrow \text{HIGH} && (\text{approaching distress threshold}) \\ |\Delta T| \geq 23^{\circ}\text{C} &\rightarrow \text{CRITICAL} && (\text{IRPWM §7 distressing required}) \end{aligned}$$

Failure mode:

$$\begin{aligned} \Delta T > 0 &\rightarrow \sigma_{\text{th}} > 0 \text{ (compressive)} &\rightarrow \text{CWR Buckling risk} \\ \Delta T < 0 &\rightarrow \sigma_{\text{th}} < 0 \text{ (tensile)} &\rightarrow \text{Transverse Fracture risk} \end{aligned}$$

Buckling threshold (approximate):

$$\sigma_{\text{th}} \approx +72.5 \text{ MPa corresponds to } \Delta T = +30^{\circ}\text{C}$$

Fracture threshold (approximate):

$$\sigma_{th} \approx -48.3 \text{ MPa corresponds to } \Delta T = -20^{\circ}\text{C}$$

Contribution to total stress range (for Paris Law):

When a USFD crack is present, thermal stress also contributes to the Mode I stress intensity factor, especially for transverse cracks. RHIE adds 30% of the absolute thermal stress to the bending $\Delta\sigma$:

$$\Delta\sigma_{total} = \Delta\sigma_{bending} + 0.30 \times |\sigma_{th}| \quad [\text{MPa}]$$

The 0.30 factor reflects that thermal stress contributes to crack opening (Mode I) primarily through rail restraint relaxation effects. This is conservative for the Paris Law calculation.

Why thermal stress is NOT in the Health Index:

Thermal stress is seasonal and reversible — it goes away at night, in cooler seasons, and immediately after destressing. Including it in the health score would make the health index oscillate with the weather rather than reflect the actual structural degradation of the rail. Thermal risk is shown as a separate tag (LOW/MODERATE/HIGH/CRITICAL) and drives TSR and destressing recommendations independently.

SFT values by climate zone:

Zone ID	Zone Name	SFT (°C)	Rationale
1	Thar Desert	35	High ambient, rail lays hot
2	Semi-Arid Rajasthan	31	Moderate-high ambient
3	Punjab–Haryana Plains	25	Cold winters, hot summers
4	Upper Gangetic Plains (Delhi)	28	Similar to Punjab
5	Middle Gangetic Plains	27	Slightly higher humidity
6	Eastern Gangetic Plains	27	Bihar flood plains
7	Bengal Delta	28	Maritime influence
8	North Bengal Terai	25	Monsoon heavy
9	Brahmaputra Valley	24	Coolest, extreme monsoon
10	Northeast Hills	20	Highest elevation, coolest
11	CG–Jharkhand Plateau	28	Hot dry summers
12	MP–Bundelkhand	28	Plateau, dry
13	Vidarbha	30	Hottest Indian zone, 44°C peak
14	Interior Maharashtra	28	Deccan Plateau
15	Konkan–Coastal Maharashtra	28	Heavy rainfall, moderate temp

Zone ID	Zone Name	SFT (°C)	Rationale
16	Andhra–Telangana Interior	29	Hot Deccan
17	Coastal AP–Odisha	29	Coastal, cyclone risk
18	TN–Karnataka Coast	29	NE Monsoon region
19	Kerala–Western Ghats	28	Extreme SW Monsoon
20	Gujarat–Saurashtra	30	Hot, Kutch cyclone risk

6.4 Paris Law Fatigue Crack Propagation

Purpose: Given a USFD-detected crack of initial depth a_0 (mm), compute the crack growth rate (da/dN per cycle), the critical crack depth (a_c) at which K_{IC} is reached and fracture occurs, and the Remaining Useful Life (RUL) in days.

Reference: Paris and Erdogan (1963), applied to rail steel by Zerbst et al. (2005), calibrated with $K_{IC} = 29 \text{ MPa}\sqrt{\text{m}}$ for GR1080.

Derivation:

Step 1: Geometry correction factor
 $Y = 1.12 + 0.08 \times \text{curvature_degrees}$
 ($Y = 1.12$ for tangent track; increases on curves due to higher stress concentrations)

Step 2: Stress intensity factor range
 $\Delta K = Y \times \Delta\sigma \times \sqrt{(\pi \times a_0)}$ [MPa $\sqrt{\text{m}}$]
 (a_0 in metres, $\Delta\sigma = \Delta\sigma_{\text{bending}} + 0.30 \times |\sigma_{\text{th}}|$)

Step 3: Crack growth rate per cycle
 $da/dN = C \times (\Delta K)^m$ [m/cycle]
 $da/dN = 1.65 \times 10^{-11} \times (\Delta K)^3$

Step 4: Critical crack depth (fracture condition $K = K_{IC}$)
 $K_{IC} = Y \times \Delta\sigma \times \sqrt{(\pi \times a_c)}$
 $a_c = (K_{IC} / (Y \times \Delta\sigma))^2 / \pi$ [m]

Step 5: Remaining cycles to fracture (analytical Paris integral for $m=3$)
 $N_r = (2/\text{den}) \times (1/\sqrt{a_0} - 1/\sqrt{a_c})$
 where $\text{den} = C \times (Y \times \Delta\sigma)^3 \times \pi^{1.5}$

Step 6: Remaining Useful Life
 $\text{RUL} = N_r / \text{estFreq}$ [days]
 (estFreq = axle passes per day from route class)

Step 7: Daily crack growth rate
 $da/dt = da/dN \times \text{estFreq} \times 10^6$ [$\mu\text{m}/\text{day}$]

Crack trajectory for visualisation:

The dashboard numerically integrates Paris Law to produce a crack growth curve (crack depth vs. days) using adaptive time steps:

For each time step dt :
 $a_{\text{new}} = a + C \times (Y \times \Delta\sigma \times \sqrt{(\pi \times a)})^3 \times \text{estFreq} \times dt$

This generates the trajectory chart in the Health Analysis tab showing crack depth vs. days until the critical depth a_c is reached.

Example — 2mm crack, Route A, 60 kg/m, Delhi (June):

- $\Delta\sigma_{\text{bending}} = 126.4 \text{ MPa}$ (from Winkler)
- $\sigma_{\text{th}} = 210,000 \times 11.5 \times 10^{-6} \times (66-28) = 91.8 \text{ MPa} \rightarrow \text{contributes } 0.30 \times 91.8 = 27.5 \text{ MPa}$
- $\Delta\sigma_{\text{total}} = 126.4 + 27.5 = 153.9 \text{ MPa}$
- $Y = 1.12$ (tangent track)
- $\Delta K = 1.12 \times 153.9 \times \sqrt{(\pi \times 0.002)} = 1.12 \times 153.9 \times 0.0792 = 13.65 \text{ MPa}\sqrt{\text{m}}$
- $a_c = (29 / (1.12 \times 153.9))^2 / \pi = (0.168)^2 / \pi = 0.00898 \text{ m} = 8.98 \text{ mm}$
- $da/dN = 1.65 \times 10^{-11} \times 13.65^3 = 4.19 \times 10^{-8} \text{ m/cycle} = 41.9 \text{ nm/cycle}$
- $\text{estFreq (Route A)} = 20 \times 86 + 40 \times 96 = 1720 + 3840 = 5560 \text{ axle passes/day}$
- $N_r = (2/\text{den}) \times (1/\sqrt{0.002} - 1/\sqrt{0.00898})$
- $\text{RUL} \approx 130 \text{ days}$

Note on RUL interpretation: RUL is used exclusively for defect alerting (red/amber/green flags, TSR recommendation, inspection schedule). It is **NOT** included in the health score. A 2mm crack with 130-day RUL on a young track should not drag the health score to 40/100 — the structural condition of the track is otherwise good.

6.5 GMT Wear Model

Purpose: Compute rail head vertical wear as a function of cumulative gross million tonnes (GMT) and curvature.

Reference: Nippon Steel K003en track test data (DHH370 rail, R=290m curve, BG equivalent); calibrated with RDSO field measurements.

Wear rates:

Track Geometry	Rate	Condition
Tangent (curvature = 0°)	0.013 mm/MGT	Standard BG main line
Mild curve (curvature < 1°)	0.025 mm/MGT	~R > 1600m
Sharp curve (curvature ≥ 1°)	0.044 mm/MGT	~R < 1600m, high lateral forces

Wear (mm) = GMT_cumulative × rate
Wear fraction = Wear / Renewal_limit

The timeline chart projects both GMT accumulation and wear to the renewal date using the constant annual GMT rate from the route class.

7. Indian Climate Zones — All 20

Each zone is defined by: zone ID, name, states covered, SFT (°C), annual rainfall (mm), seismic zone (BIS 1893), terrain type, monsoon risk level, 12-month ambient temperature array (°C), and 12-month solar gain array (°C above ambient).

Rail temperature = ambient[month] + solar_gain[month]

The solar gain represents the additional temperature rise due to direct solar radiation on the dark steel rail head, typically 10–30°C above ambient in peak summer.

ID	Zone Name	States	SFT°C	Rain mm	Seismic	Terrain	Monsoon Risk
1	Thar Desert	W. Rajasthan	35	150	III	Desert	Low
2	Semi-Arid Rajasthan	E. Rajasthan	31	400	II	Semi-Arid	Moderate
3	Punjab–Haryana Plains	Punjab, Haryana	25	580	IV	Alluvial	Moderate
4	Upper Gangetic Plains	Delhi, W. UP	28	600	IV	Alluvial	Moderate
5	Middle Gangetic Plains	E. UP, Lucknow	27	850	III	Alluvial	High
6	Eastern Gangetic Plains	E. UP, Bihar	27	1050	III	Flood Plains	Very High
7	Bengal Delta	S. West Bengal	28	1600	III	Delta/Coastal	Very High (Cyclone)
8	North Bengal Terai	N. Bengal	25	2500	IV	Terai/Flood	Extreme
9	Brahmaputra Valley	Assam Plains	24	2800	V	Flood/Embankment	Extreme
10	Northeast Hills	Meghalaya, Manipur	20	11000	V	Hill/Seismic	Extreme + Landslide
11	CG–Jharkhand Plateau	CG, Jharkhand	28	1300	II	Plateau	High
12	MP–Bundelkhand	Madhya Pradesh	28	900	II	Plateau/Hilly	Moderate
13	Vidarbha	Nagpur, Amravati	30	1050	II	Deccan Plateau	Moderate

ID	Zone Name	States	SFT°C	Rain mm	Seismic	Terrain	Monsoon Risk
14	Interior Maharashtra	Pune, Nashik, Aurangabad	28	700	III	Deccan Plateau	Moderate
15	Konkan–Coastal Mah.	Mumbai, Goa	28	4000	III	Coastal/Hilly	Very High (Flooding)
16	Andhra–Telangana Interior	Hyderabad, Warangal	29	900	II	Deccan Plateau	Moderate
17	Coastal AP–Odisha	Vizag, Vijayawada, BBSR	29	1200	II	Coastal/Delta	High (Cyclone)
18	TN–Karnataka Coast	Chennai, Bengaluru, Mysuru	29	1100	II	Coastal Plateau	Moderate (NE Monsoon)
19	Kerala–Western Ghats	Kerala, Coastal Karnataka	28	3200	III	Coastal/Ghats	Extreme (SW Monsoon)
20	Gujarat–Saurashtra	Gujarat, Kutch	30	550	III	Coastal/Arid	Moderate (Cyclone-Kutch)

Monthly Ambient Temperature (°C) — All 20 Zones:

Zone	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1 Thar Desert	8	12	20	31	40	44	38	36	33	28	18	10
2 Semi-Arid Raj	10	14	22	32	40	42	36	34	32	26	16	11
3 Punjab–Haryana	6	10	18	28	38	40	36	35	32	24	13	7
4 Upper Ganges	7	11	19	30	40	42	38	36	33	26	14	8
5 Middl	10	14	22	32	38	40	36	35	33	28	16	11

Zone	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
e Gange tic												
6 E. Gange tic	12	16	24	33	38	38	34	34	33	29	18	13
7 Benga l Delta	14	18	26	33	36	35	32	32	32	30	22	15
8 N. Benga l Terai	10	13	20	28	30	30	28	28	28	24	16	11
9 Brah maput ra	12	15	20	26	28	30	28	28	27	24	16	12
10 NE Hills	8	10	16	20	22	24	24	24	22	18	12	8
11 CG– Jhark hand	14	18	26	34	40	38	32	32	32	28	18	14
12 MP– Bunde lkhan d	12	16	24	34	40	40	34	33	32	26	16	12
13 Vidar bha	16	20	28	36	44	42	33	32	32	28	20	16
14 Interi or Mah	14	17	24	32	38	34	29	28	28	26	18	14
15 Konk an	20	22	28	32	35	32	29	29	29	30	26	22
16 AP – Telan gana	18	22	28	34	38	36	30	29	29	27	21	18
17 Coast	22	25	30	34	36	36	32	32	32	29	24	22

Zone	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
al AP												
18 TN — Karna taka	24	26	30	33	36	36	34	34	33	30	27	24
19 Keral a— W.Gh ats	24	26	28	30	32	28	26	26	27	28	26	24
20 Gujar at	16	20	26	32	38	38	34	32	32	28	20	16

Monthly Solar Gain (°C above ambient) — All 20 Zones:

Zone	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1 Thar Desert	10	12	20	25	28	30	14	14	22	20	12	9
2 Semi- Arid Raj	9	11	18	22	24	26	12	12	20	18	11	8
3 Punja b— Harya na	8	10	14	18	20	22	10	10	16	14	9	7
4 Upper Gange tic	9	11	16	20	22	24	11	11	18	16	10	8
5 Middl e Gange tic	8	10	14	18	20	22	10	10	16	14	9	7
6 E. Gange tic	7	9	13	16	18	10	6	6	11	12	8	6
7 Benga l Delta	6	8	12	16	18	8	5	5	10	12	8	6

Zone	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
8 N. Bengal Terai	5	7	9	12	12	6	4	4	8	9	6	5
9 Brahmaputra	4	6	8	10	10	4	3	3	6	8	5	4
10 NE Hills	3	4	6	8	8	3	2	2	5	6	4	3
11 CG–Jharkhand	9	11	15	19	21	10	6	6	12	13	9	7
12 MP–Bundelkhand	9	11	16	20	22	22	10	10	16	15	10	8
13 Vidarbha	9	11	16	20	23	18	8	8	12	14	9	8
14 Interior Mah	8	10	14	18	20	12	6	6	10	12	8	7
15 Konk an	6	8	10	14	16	6	3	3	6	8	6	5
16 AP – Telan gana	8	10	14	18	20	10	6	6	10	12	8	7
17 Coastal AP	7	9	13	17	18	9	5	5	9	10	7	6
18 TN – Karna taka	8	10	14	17	17	8	5	5	9	10	7	6
19 Kerala–W.Gh ats	6	7	10	12	14	4	3	3	5	6	5	5

Zone	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
20 Gujar at	9	11	15	20	22	20	10	10	16	16	10	8

8. Indian Railways Zones and Division–Climate Mapping

18 Zonal Railways, 68 Divisions. Each division is mapped to one of the 20 climate zones. The division selection auto-populates all climate-specific parameters (SFT, rainfall, seismicity, terrain, monthly temperatures).

IR Zone	Full Name	HQ	Divisions → Climate Zone ID
CR	Central Railway	Mumbai	Mumbai(CR) → 15, Pune → 14, Solapur → 16, Bhusawal → 12, Nagpur(CR) → 13
ER	Eastern Railway	Kolkata	Howrah → 7, Sealdah → 7, Asansol → 11, Malda → 8
ECR	East Central Railway	Hajipur	Hajipur → 6, Samastipur → 6, Danapur → 6, Dhanbad → 11, DDU(Mughalsarai) → 5
ECoR	East Coast Railway	Bhubaneswar	Khurda Road → 17, Sambalpur → 11, Visakhapatnam → 17
NR	Northern Railway	New Delhi	Delhi → 4, Ambala → 3, Moradabad → 5, Lucknow(NR) → 5, Firozpur → 3
NCR	North Central Railway	Prayagraj	Prayagraj → 5, Agra → 4, Jhansi → 12
NER	North Eastern Railway	Gorakhpur	Lucknow(NER) → 5, Varanasi → 5, Izzatnagar → 5
NFR	Northeast Frontier Railway	Guwahati	Alipurduar → 8, Katihar → 8, Rangiya → 9, Lumding → 9, Tinsukia → 10

IR Zone	Full Name	HQ	Divisions → Climate Zone ID
NWR	North Western Railway	Jaipur	Jaipur → 2, Ajmer → 2, Bikaner → 1, Jodhpur → 1
SR	Southern Railway	Chennai	Chennai → 18, Madurai → 18, Salem → 18, Tiruchirapalli → 18, Thiruvananthapuram → 19, Palakkad → 19
SCR	South Central Railway	Secunderabad	Secunderabad → 16, Hyderabad → 16, Guntakal → 16, Guntur → 17, Vijayawada → 17, Nanded → 13
SECR	South East Central Railway	Bilaspur	Bilaspur → 11, Raipur → 11, Nagpur(SECR) → 13
SER	South Eastern Railway	Kolkata	Adra → 11, Chakradharpur → 11, Kharagpur → 7, Ranchi → 11
SWR	South Western Railway	Hubballi	Hubballi → 14, Bengaluru → 18, Mysuru → 18
WR	Western Railway	Mumbai	Mumbai(WR) → 15, Vadodara → 20, Ahmedabad → 20, Rajkot → 20, Bhavnagar → 20, Ratlam → 12
WCR	West Central Railway	Jabalpur	Jabalpur → 12, Bhopal → 12, Kota → 2

9. Seasonal Calendar

Five seasons are defined per the Indian meteorological calendar. Each season has an associated CWR risk profile:

Season	Months (0=Jan)	Months (Names)	CWR Risk
Winter	11, 0, 1	Dec, Jan, Feb	Tensile stress → Transverse Fracture risk

Season	Months (0=Jan)	Months (Names)	CWR Risk
Spring	2, 3	Mar, Apr	Destressing Window (optimal time for CWR destressing)
Summer	4, 5	May, Jun	Compressive peak → CWR Buckling (peak solar gain)
Monsoon	6, 7, 8	Jul, Aug, Sep	Ballast Wash / Mud Pumping risk
Autumn	9, 10	Oct, Nov	Post-Monsoon Geometry degradation

10. Health Index — Formula and Scoring

Philosophy

The Health Index (0–100) measures the **structural and maintenance condition** of the track segment at the time of assessment. It uses a **deduction-based system**: start at 100, subtract penalties from six independent factors.

Key design decisions:

- **Thermal stress excluded:** Seasonal, reversible. A hot June rail in Rajasthan is not permanently damaged — it restores overnight. Thermal risk is reported as a separate classification (LOW/MODERATE/HIGH/CRITICAL).
- **RUL excluded:** Crack RUL drives emergency alerts and TSR recommendations separately. Including it in health would collapse the score of a physically young track with a small crack detected early by USFD.
- **Crack fraction uses bending-only $\Delta\sigma$ for a_c:** So health doesn't swing with seasons (avoids $\Delta\sigma_{\text{total}}$, which includes thermal contribution).

Formula

```
Health = round( max(1, 100 - (d_gmt + d_wear + d_maint + d_ballast + d_crack + d_curv)) )
```

where:

```
d_gmt      = 40 × min(1, gmtF)                – structural, irreversible
d_wear     = 20 × min(1, wearR.frac)           – structural, irreversible
d_maint    = 15 × min(1, maint_days / 365)     – maintenance-reversible
d_ballast  = 15 × max(0, (300 - ballast_mm) / 100) – maintenance-reversible
d_crack    = 7 × (crack_mm > 0 ? min(1, crack_mm / a_c_health) : 0)
d_curv     = 3 × min(1, curvature_degrees / 5) – geometry penalty
```

Critical crack depth for health (bending only):

$$Y_h = 1.12 + 0.08 \times \text{curvature_degrees}$$

$$a_{c_health} = (K_{IC} / (Y_h \times \Delta\sigma_{\text{bending}}))^2 / \pi \quad [\text{mm}]$$

Component Weights and Interpretation

Component	Max Deduction	Drives	Type
GMT Life Fraction	40 pts	Structural degradation accumulation	Irreversible
Rail Head Wear Fraction	20 pts	Material loss at head	Irreversible
Maintenance Age	15 pts	Track geometry degradation since last tamping	Reversible (tamp restores)
Ballast Deficiency	15 pts	Load distribution / dynamic amplification	Reversible (BCM/renewal restores)
Crack Fraction vs Critical	7 pts	Fatigue structural integrity	Partially reversible (cropping)
Curvature	3 pts	Permanent geometry penalty	Irreversible
Total possible	100 pts		

Score Ranges and Interpretation

Score	Condition	Typical Stage	Action
90–100	Excellent	0 (New Rail)	Routine monitoring. Focus on preventive grinding.
75–89	Good	1 (Early Wear)	Standard maintenance programme. Tamping schedule.
55–74	Fair	2 (Active Degradation)	Intensified USFD. Corrective grinding. Renewal planning.
35–54	Poor	3 (Critical Fatigue)	Urgent intervention. TSR consideration. Renewal imminent.
1–34	Critical	4 (End of Life)	Immediate action. TSR imposed. Renewal within one block.

Example Calculations

New 60kg/m track, 2021 installation, Route A, Delhi (5 years old, no crack, standard ballast):

- $\text{gmtF} = (35 \times 5) / 800 = 0.219 \rightarrow \text{d_gmt} = 40 \times 0.219 = 8.8$
- $\text{wearR.frac} = (175 \times 0.013) / 8 = 0.284 \rightarrow \text{d_wear} = 20 \times 0.284 = 5.7$
- $\text{maint} = 90 \text{ days} \rightarrow \text{d_maint} = 15 \times 0.247 = 3.7$
- $\text{ballast} = 280 \text{ mm} \rightarrow \text{d_ballast} = 15 \times (20/100) = 3.0$
- $\text{crack} = 0 \rightarrow \text{d_crack} = 0$
- $\text{curv} = 0 \rightarrow \text{d_curv} = 0$
- **Health = 100 – (8.8+5.7+3.7+3.0) = 79**

Old 52kg/m track, 1998 installation, Route B, 2mm crack, overdue maintenance:

- $\text{gmtF} = (8 \times 25) / 525 = 0.381 \rightarrow \text{d_gmt} = 40 \times 0.381 = 15.2$
- $\text{wear} = 200 \times 0.013 = 2.6\text{mm}$, $\text{frac} = 2.6/13 = 0.20 \rightarrow \text{d_wear} = 4.0$
- $\text{maint} = 240 \text{ days} \rightarrow \text{d_maint} = 15 \times 0.658 = 9.9$
- $\text{ballast} = 260 \text{ mm} \rightarrow \text{d_ballast} = 15 \times 0.40 = 6.0$
- $\text{crack} = 2\text{mm}$, $\text{a_c} \approx 25\text{mm}$ (52kg, lower $\Delta\sigma$) $\rightarrow \text{crackFrac} = 0.08 \rightarrow \text{d_crack} = 0.56$
- $\text{curv} = 2^\circ \rightarrow \text{d_curv} = 3 \times 0.40 = 1.2$
- **Health = 100 – (15.2+4.0+9.9+6.0+0.56+1.2) = 63**

11. Rail Stage Classification

Stage is determined purely by structural degradation — GMT fraction and crack fraction. Thermal risk does not determine stage (it's a risk overlay, not a structural stage).

```
if crackF > 0.70 OR gmtF > 0.90 → Stage 4: End of Life
if crackF > 0.40 OR gmtF > 0.70 → Stage 3: Critical Fatigue
if crackF > 0.15 OR gmtF > 0.40 → Stage 2: Active Degradation
if gmtF > 0.12 → Stage 1: Early Wear
else → Stage 0: New Rail
```

where:

- $\text{gmtF} = \text{GMT_cumulative} / \text{GMT_limit}$
- $\text{crackF} = \text{crack_mm} / \text{a_c_mm}$ (using bending $\Delta\sigma$ only for a_c)

Stage	Label	Color	Description
0	New Rail	#4D9B60 (green)	Within initial wear-in period. Surface fatigue not initiated. Preventive grinding is most effective now.
1	Early Wear	#7AAD72 (light green)	Corrugation onset. Surface fatigue initiated at rail head.

Stage	Label	Color	Description
			Preventive grinding and tamping are the governing interventions.
2	Active Degradation	#C8822A (amber)	RCF developing. Gauge corner wear on curves. Mandatory USFD monitoring. Corrective grinding and re-profiling required.
3	Critical Fatigue	#B03030 (red)	USFD-detectable cracks, shelling, or spalling present. Immediate intervention required. Cropping before renewal.
4	End of Life	#7A0020 (dark red)	GMT limit reached or crack approaching critical depth. Renewal must be planned within next possession block.

12. ML Model — Complete Specification

12.1 Architecture and Training

RHIE uses **Gradient Boosting** for two simultaneous tasks:

1. **Regression:** Predict remaining useful life in MGT (how many gross million tonnes remain before the rail needs renewal)
2. **Classification:** Classify the dominant defect type from 5 classes

The in-browser model (training during page load) uses a custom JavaScript implementation. The Python validation model (used to generate `RHIE_ML_metrics.csv`) uses scikit-learn's `GradientBoostingRegressor` and `GradientBoostingClassifier`.

Gradient Boosting algorithm:

Initialize: $F_0(x) = \text{mean}(y)$ [base prediction]

For $t = 1$ to T :

1. Compute residuals: $r_i = y_i - F_{t-1}(x_i)$
2. Fit regression tree h_t to residuals
3. Update: $F_t(x) = F_{t-1}(x) + \eta \times h_t(x)$
(η = learning rate)

Final prediction: $F(x) = F_0 + \sum_i \eta \times h_i(x)$

Each tree corrects the errors of the ensemble so far. With a small learning rate (0.08) and many trees (200), the model gradually fits the training data without overfitting.

12.2 Feature Engineering

10 input features per training sample:

#	Feature Name	Data Type	Source	Range (typical)
1	GMT Accumulated (MGT)	Continuous	Physics: route class $\times$ age	0–1100 MGT
2	Zone ID (1–20)	Integer	Climate zone selection	1–20
3	Rail Type (0=52kg, 1=60kg)	Binary	User input	0 or 1
4	Axle Load (t)	Continuous	User input	14–32 t
5	Curvature (°)	Continuous	User input	0–5°
6	Crack Depth (mm)	Continuous	USFD inspection input	0–10 mm
7	Ballast Deficit (mm)	Continuous	300mm spec minus actual	0–100 mm
8	Maintenance Age (days)	Continuous	Days since last tamping	0–400 days
9	Thermal Deviation (°C)	Continuous	Physics: $T_{\text{rail}} - \text{SFT}$	–25 to +45°C
10	Wear Depth (mm)	Continuous	Physics: GMT $\times$ wear rate	0–10 mm

12.3 Target Variables

Regression target: $\text{RUL_MGT} = \max(0, \text{GMT_limit} - \text{GMT_accumulated} + \text{noise})$

This is the residual GMT life — how many more MGT the rail can carry before reaching its renewal limit. Gaussian noise ($\sigma=15$ MGT) is added to simulate real-world variability in wear rate and maintenance history.

Classification target (5 classes):

Class	Label	Primary Trigger	Secondary
0	No Defect	GMT < 30% and no crack and moderate thermal	—
1	Corrugation	GMT > 10% (early onset)	Not curved, not crack
2	RCF / Head Wear	GMT > 30% AND curvature > 1.5°	Gauge corner fatigue

Class	Label	Primary Trigger	Secondary
3	Fatigue Crack / Life Limit	GMT > 55% OR crack > 5mm	Near end-of-life
4	CWR Thermal Risk		ΔT

4% label noise is injected to simulate USFD misclassification and borderline cases.

12.4 Training Data Generation

Python validation model (1500 samples, balanced):

- 300 samples per class, generated with class-specific feature distributions
- Class 0: early life (GMT < 30% limit), small thermal deviation
- Class 1: mid-life (GMT 10–60%), any curvature, mild thermal
- Class 2: mid-to-late life (GMT 30–85%), forced curvature > 1.5°, crack 0.5–3mm
- Class 3: late life (GMT 55–105%), crack 2–10mm
- Class 4: any GMT, forced $|\Delta T| > 25^\circ\text{C}$

Browser model (500 samples, imbalanced, organic):

- Organically generated from random parameter draws
- Defect class assigned by deterministic physics rules with 5% noise
- Class distribution approximately: [35% No Defect, 35% Corrugation, 10% RCF, 10% Fatigue, 10% CWR Thermal]
- Less balanced than the Python model but faster to generate

12.5 Hyperparameters

Parameter	Browser (JS)	Python (scikit-learn)
Algorithm	Custom JS GradientBoosting	scikit-learn GradientBoostingRegressor/Classifier
n_estimators	30	200
max_depth	3	4
learning_rate	0.15	0.08
subsample	1.0 (full)	0.8 (stochastic)
min_samples_leaf	4	5
n_splits (CV)	—	5-fold KFold
Training samples	400 (80% of 500)	1200 (80% of 1500)
Validation samples	100	300
Random seed	—	42

The Python model is stronger (more trees, stochastic subsampling, larger balanced dataset). The browser model is intentionally lightweight to complete training in ~3 seconds.

12.6 Training Curve

RMSE on held-out test set vs. number of boosting iterations (Python scikit-learn, 200 trees):

Iteration	Train RMSE (MGT)	Test RMSE (MGT)
1	189.04	194.33
5	138.39	142.10
10	93.26	95.93
15	63.11	65.10
20	43.56	45.28
25	31.02	32.93
30	23.37	25.60
40	16.31	19.22
50	14.12	17.57
75	12.38	16.84
100	11.38	16.86
125	10.59	16.89
150	10.04	17.01
175	9.38	17.06
200	8.69	17.12

Observation: Train RMSE continues to fall after iteration 75 while test RMSE plateaus around 16–17 MGT. This is mild overfitting — the optimal stopping point for test performance is around iteration 60–80. The final model at 200 trees is deployed since the overfitting is modest (0.5 MGT gap) and synthetic data noise is the limiting factor.

12.7 Regression Metrics — RUL Prediction

Metric	Test Value	5-Fold CV Mean	5-Fold CV Std	Unit
RMSE	17.12	17.33	0.997	MGT
MAE	13.65	—	—	MGT
R ²	0.9933	0.9929	0.0007	—

R² = 0.9933: The model explains 99.33% of the variance in RUL. This is high because the dominant driver (GMT accumulated) is directly available as a feature, making the regression task relatively well-determined.

RMSE = 17.12 MGT: On average, the model's RUL prediction is off by ±17 MGT. For a 800 MGT rail life, this is a ±2.1% error — adequate for maintenance scheduling purposes (typically

plan in 6–12 month windows).

Cross-validation stability: $CV R^2 = 0.9929 \pm 0.0007$ across 5 folds — very low variance, confirming the model generalises well and is not overfit to any particular fold.

MAE = 13.65 MGT: Mean Absolute Error is lower than RMSE, indicating the error distribution has a few larger outliers pulling RMSE up, but the median prediction is more accurate.

12.8 Classification Metrics — Defect Type

Metric	Test Value	5-Fold CV Mean	5-Fold CV Std
Accuracy	0.9367	0.9453	0.0177
Weighted Precision	0.9376	—	—
Weighted Recall	0.9367	—	—
Weighted F1-Score	0.9370	0.9451	0.0177
Macro Precision	0.9376	—	—
Macro Recall	0.9367	—	—
Macro F1-Score	0.9370	—	—

12.9 Per-Class Performance

Class	Precision	Recall	F1-Score	Support
No Defect	0.887	0.917	0.902	60
Corrugation	0.869	0.883	0.876	60
RCF / Head Wear	0.949	0.933	0.941	60
Fatigue Crack / Life Limit	0.983	0.950	0.966	60
CWR Thermal Risk	1.000	1.000	1.000	60

CWR Thermal Risk (F1=1.000): Perfect classification. The thermal deviation feature is a dominant, clean signal — when $|\Delta T| > 23^\circ\text{C}$, the model assigns CWR Thermal Risk with 100% accuracy. No other class has comparable thermal signatures.

Fatigue Crack / Life Limit (F1=0.966): Near-perfect. Crack depth and GMT accumulated together cleanly separate this class from others.

RCF / Head Wear (F1=0.941): Strong. The curvature + GMT combination discriminates well.

No Defect and Corrugation (F1≈0.90, 0.88): The main confusion is between these two classes — 5 No Defect samples classified as Corrugation, 7 Corrugation classified as No Defect. This is expected: early corrugation is hard to distinguish from "no defect" without direct rail profile measurement.

12.10 Confusion Matrix

Rows = Actual class, Columns = Predicted class:

Actual \ Predicted	No Defect	Corrugation	RCF/Head Wear	Fatigue Crack	CWR Thermal
No Defect	55	5	0	0	0
Corrugation	7	53	0	0	0
RCF / Head Wear	0	3	56	1	0
Fatigue Crack / Life Limit	0	0	3	57	0
CWR Thermal Risk	0	0	0	0	60

Key observations:

- All errors are concentrated in the No Defect ↔ Corrugation boundary (12 total misclassifications)
- RCF/Head Wear is sometimes confused with Corrugation (3) or Fatigue Crack (1) — both neighbouring stages
- Zero cross-class errors between thermal and structural defects — the thermal feature completely separates Class 4
- No false negatives on CWR Thermal Risk — no CWR thermal cases missed

12.11 Feature Importance

Regression (RUL prediction) — by impurity gain:

Rank	Feature	Importance
1	GMT Accumulated (MGT)	0.6758 (67.6%)
2	Rail Type (0=52kg, 1=60kg)	0.2793 (27.9%)
3	Crack Depth (mm)	0.0413 (4.1%)
4	Wear Depth (mm)	0.0007
5	Maintenance Age (days)	0.0006
6	Axle Load (t)	0.0006
7	Ballast Deficit (mm)	0.0006
8	Thermal Deviation (°C)	0.0004
9	Curvature (°)	0.0003
10	Zone ID	0.0003

GMT (67.6%) and Rail Type (27.9%) dominate because the RUL formula is primarily $\max(0,$

GMT_limit - GMT_accumulated) where GMT_limit depends on rail type (800 for 60kg, 525 for 52kg). This is physically correct — the most important predictor of how much life is left is how much has been consumed.

Classification (defect type) — by impurity gain:

Rank	Feature	Importance
1	Crack Depth (mm)	0.4393 (43.9%)
2	Thermal Deviation (°C)	0.3392 (33.9%)
3	Wear Depth (mm)	0.0958 (9.6%)
4	GMT Accumulated (MGT)	0.0712 (7.1%)
5	Curvature (°)	0.0318 (3.2%)
6	Rail Type	0.0103
7	Ballast Deficit (mm)	0.0053
8	Zone ID	0.0030
9	Maintenance Age (days)	0.0021
10	Axle Load (t)	0.0020

For classification, crack depth (44%) and thermal deviation (34%) dominate — these are the most discriminating signals. Crack separates Fatigue class from others; thermal deviation uniquely identifies CWR Thermal Risk. Wear depth (10%) helps separate RCF from Corrugation. GMT (7%) captures life-stage effects. Curvature (3%) separates RCF from corrugation.

12.12 What Each Metric Means

RMSE (Root Mean Squared Error): Square root of the average squared prediction error.

Penalises large errors more than MAE. RMSE = 17.12 MGT means on average, errors are on the order of ± 17 MGT. Since we're predicting residual life on a 0–800 MGT scale, this is a 2.1% relative error. The metric is in MGT (the same units as the prediction target).

MAE (Mean Absolute Error): Simple average of absolute errors. MAE = 13.65 MGT < RMSE = 17.12 MGT indicates the error distribution is right-skewed — most predictions are within 14 MGT of truth, but a few outliers have errors of 30–50 MGT which inflate RMSE.

R² (Coefficient of Determination): Proportion of variance in the target variable explained by the model. R² = 0.9933 means the model accounts for 99.33% of the variance in RUL. The remaining 0.67% is unexplained (random variation). R² ranges from $-\infty$ to 1.0; a value of 0 means "predict the mean" is as good as the model; 1.0 is perfect. 0.993 is very high for an engineering regression task.

Accuracy: Fraction of test samples correctly classified. Accuracy = 0.937 means 93.7% of the 300 test samples were assigned the correct defect class. With 5 balanced classes, random chance would give 20% accuracy, so the model is performing dramatically better than chance.

Precision (per class): Of all samples the model predicted as Class X, what fraction actually were Class X? High precision means few false alarms for that class. Precision = 1.000 for CWR Thermal Risk means every thermal risk prediction was correct.

Recall (per class): Of all actual Class X samples, what fraction did the model find? High recall means few missed detections. Recall = 1.000 for CWR Thermal Risk means every actual thermal risk case was detected.

F1-Score: Harmonic mean of Precision and Recall: $F1 = 2 \times P \times R / (P + R)$. Balances the two — useful when false positives and false negatives have similar costs. F1 = 1.0 is perfect; F1 = 0.0 is worst.

Weighted F1: F1 averaged across classes, weighted by the number of samples in each class (support). Since the dataset is balanced (60 per class), weighted F1 $\approx$ macro F1 $\approx$ simple average F1 here.

5-Fold Cross-Validation: The dataset is split into 5 equal folds. The model is trained on 4 folds and evaluated on the 5th, repeated 5 times rotating which fold is held out. The final CV score is the mean and standard deviation across the 5 evaluation runs. This tests that the model doesn't happen to perform well on one particular 80/20 split. CV $R^2 = 0.9929 \pm 0.0007$ confirms the model is stable and not dependent on a lucky split.

13. Defect Detection and Repair Ranking

Stage-Based Repair Protocol

The base repair list for each stage reflects standard IRPWM intervention sequencing:

Stage	Base Repair Order (METHODS IDs)
0 — New Rail	[13, 1, 3, 29] — Preventive grinding, tamping, profiling, rail anchors
1 — Early Wear	[13, 1, 14, 2, 3] — Preventive then corrective grinding, tamping
2 — Active Degradation	[14, 15, 1, 2, 47, 13] — Corrective grinding, milling, tamping, lubrication
3 — Critical Fatigue	[20, 14, 1, 26, 13, 23] — Cropping, grinding, tamping, distress, stitching
4 — End of Life	[48, 17, 4, 5, 36, 14] — Full renewal, FBW, ballast cleaning, sleeper renewal

Physics-Condition Additions

Repairs are prepended/appended based on physics findings:

Condition	Trigger	Added Repairs
Significant crack	crack > 3mm	[21, 20] — Crack tip hole + Cropping (prepend)
Detected crack	crack > 0.5mm	[20, 23] — Cropping + Stitching (prepend)

Condition	Trigger	Added Repairs
CWR thermal CRITICAL/HIGH	thermalRisk $\in$ {CRITICAL, HIGH}	[26, 30] — Destressing + Neutral temp adjustment
Low ballast	ballast < 270mm	[5, 4] — Deep ballast renewal + BCM
Overdue maintenance	maint > 180d	[1, 2] — Tamping + CSM/DTS pass
Tight curve	curv > 1.5°	[47, 39] — Lubrication + Check rail
GMT > 85%	gmtF > 0.85	[48] — Full P-Way renewal

Deduplication: The top 6 unique repairs are extracted in priority order (physics conditions first, then stage base list) after removing duplicates.

Governing Conditions Ranking

In the Defect & Repair tab, conditions are ranked by their computed impact score (0–1):

Condition	Impact Score	Severity Tags
GMT Life Consumed	gmtF	>0.8 → CRITICAL, >0.5 → HIGH, else OK
Crack Fraction vs Critical	crackF	>0.6 → CRITICAL, >0.3 → HIGH, else OK
CWR Thermal Stress	min(1,	ΔT
Ballast Deficiency	max(0, (300–ballast)/100)	<250 → HIGH, <280 → MODERATE
Maintenance Age	min(1, maint/365)	>270d → HIGH, >180d → MODERATE
Curvature	min(1, curv/5)	>1.5° → MODERATE

14. All 48 IRPWM Repair Methods

Category: Geometry (1–12)

ID	Method	Urgency	IRPWM Reference	Description
1	Multi-Tie Tamping	Routine	Para 220-222	Restores level, alignment, cross-level. Core scheduled maintenance operation on all routes.

ID	Method	Urgency	IRPWM Reference	Description
2	Continuous Auto Tamping (CSM+DTS)	Routine	Para 220	Continuous tamping + Dynamic Stabiliser. Mandatory on A+/HM. Immediate track stability.
3	Ballast Profiling	Routine	Para 214	Corrects shoulder width and cress profile post-tamping.
4	Ballast Cleaning (BCM)	Planned	Para 215-217	Screens fouled ballast. Critical for flood zones and waterlogged sections.
5	Deep Ballast Renewal	Planned	Para 218	Full 300mm BG ballast replacement when BCM is insufficient.
6	Geo-Textile Subgrade Reinforcement	Planned	RDSO GE	Prevents fines migration from subgrade. Geo-grid for embankment stability.
7	Manual Lifting and Packing	Emergency	Para 208	Emergency spot packing at acute settlements pending machine tamping.
8	Track Slewing	Routine	Para 225	Corrects lateral misalignment and versine errors.
9	Gauge Adjustment	Planned	Para 207	Corrects BG gauge deviations >±6mm via sleeper change or rail seat repositioning.
10	Cross-Level Correction (Shimming)	Routine	Para 230	Rail-seat shimming to correct cross-level and adjust

ID	Method	Urgency	IRPWM Reference	Description
				superelevation.
11	Drainage Rehabilitation	Planned	Para 120-125	Catch pits, cess drains, French drains, culverts. Critical Zones 7-9, 15, 19.
12	Dynamic Track Stabilisation (DTS)	Post-Tamp	Para 222A	Controlled vibration post-tamping accelerates ballast consolidation.

Category: Rail (13–25)

ID	Method	Urgency	IRPWM Reference	Description
13	Preventive Rail Grinding	Routine	RDSO/2018/Track-BD/01	Removes surface fatigue layer before crack propagation. Extends life to 1000 GMT. Every 30-50 MGT on A+.
14	Corrective Rail Grinding	Planned	RDSO/2018/Track-BD/01	Deep grinding removes corrugation and shells. 0.3-0.8mm metal removal; multiple passes.
15	Rail Head Reprofile (Milling)	Planned	RDSO	Precision reprofiles to standard cross-section. More metal removed than grinding.
16	Aluminothermic (Thermite) Welding	Emergency	Para 264-269	In-situ thermite welding. Post-heat treatment mandatory for GR1080.
17	Flash Butt Welding	Planned	Para 256-263	High-quality FBW. Standard for CWR panels and new rail.

ID	Method	Urgency	IRPWM Reference	Description
				Preferred over thermite.
18	Gas Pressure Welding	Emergency	Para 270	Temporary repair using gas heating. USFD verification post-weld required.
19	Manual Arc Repair Welding	Emergency	Para 275	Arc welding for localised surface defects. Pre-heat $\geq 200^{\circ}\text{C}$ for GR1080.
20	Cropping and Rail Insertion	Planned	Para 277	Removes defective section (min 6m) and inserts new rail with two welds.
21	Crack Tip Hole Drilling	EMERGENCY	Para 278	12-14mm hole arrests propagation. BUY TIME ONLY — section removal within 24h mandatory.
22	Clamp Fishplate + Rail Bond	EMERGENCY	Para 280	8-bolt clamp fishplates for broken rail. Emergency measure; TSR imposed immediately.
23	Rail Stitching	Planned	RDSO advisory	High-strength bolts bridge surface crack. Buys 1-2 MGT additional life.
24	Shot Peening of Rail Head	Preventive	RDSO experimental	Compressive residual stress retards crack initiation. Effective on curves.
25	Explosive Hardening	Planned	RDSO research	Controlled explosive cold-works rail head on tight curves.

Category: CWR (26–31)

ID	Method	Urgency	IRPWM Reference	Description
26	CWR Destressing	Seasonal	Para 318-326	Full thermal stress relief by cutting to SFT and re-welding. Mandatory $ \Delta T > 23^{\circ}\text{C}$.
27	CWR Conversion from SWR	Programme	Para 300-310	SWR → CWR conversion. Eliminates joint impact; reduces maintenance significantly.
28	SEJ Installation/Replacement	Planned	Para 328-335	Switch Expansion Joint at CWR termini and bridge ends.
29	Rail Anchor Installation (SWR)	Routine	Para 245	Anti-creep anchors for SWR per grade/curvature tables.
30	Neutral Temperature Adjustment	Seasonal	Para 318	Partial destressing in spring/autumn to restore SFT within range.
31	Insulated Rail Joint (IRJ) Renewal	Planned	Para 248	Replaces end-post, side-posts, joint bars. Critical for signalling continuity.

Category: Sleeper (32–38)

ID	Method	Urgency	IRPWM Reference	Description
32	Individual Sleeper Replacement	Spot	Para 244-246	Removal and replacement of defective PSC or wooden sleepers.
33	Elastic Rail Clip Replacement	Routine	Para 235	Damaged/missing e-clips or T-28

ID	Method	Urgency	IRPWM Reference	Description
				clips cause gauge spreading on PSC track.
34	Rail Pad Renewal	Planned	Para 236	Rubber/composite pads under rail. Deteriorated pads increase dynamic forces.
35	Shoulder Bolt + Insert Replacement	Routine	Para 237	HDPE inserts anchoring clips in PSC sleepers. Worn inserts allow gauge spread.
36	Through Sleeper Renewal	Programme	Para 241-243	Zone-wise wooden/steel → PSC replacement. Largest track renewal category.
37	Anti-Vibration Pad Installation	Planned	RDSO spec	Resilient pads for vibration mitigation in urban/structure-adjacent zones.
38	Wooden Sleeper Creosote Treatment	Programme	Para 239	Pressure creosote treatment extends timber life against decay and termite attack.

Category: Structural (39–48)

ID	Method	Urgency	IRPWM Reference	Description
39	Check Rail Installation/Renewal	Planned	Para 406-408	Inside main rail on curves $R < 200\text{m}$ to prevent wheel climb. Renewal when check gauge $> 44\text{mm}$.
40	Guard Rail (Bridge/Crossing)	Planned	Para 403-405	Mandatory on spans $> 60\text{m}$ to contain derailed vehicles.
41	Mn-Steel Crossing	Planned	Para 380-385	Manganese steel

ID	Method	Urgency	IRPWM Reference	Description
	Renewal			crossing renewal at 30-50 MGT. Work-hardens to 350-500 HBW.
42	S&C Full Renewal	Programme	Para 360-375	Complete turnout assembly replacement. Life 350-450 MGT Group A.
43	Subgrade Lime/Cement Stabilisation	Planned	RDSO GE	Pressure injection when tamping gives diminishing returns (formation failure).
44	Embankment Shotcrete Stabilisation	Emergency	Para 130-135	Sprayed concrete on slopes. Critical Zones 9, 10, 19.
45	Cathodic Protection (Coastal)	Planned	RDSO corrosion	Impressed current protection for Zones 7, 15, 17, 19, 20.
46	Rail Bond Testing + Re-Bonding	Routine	Para 250	>10mΩ triggers re-bonding. Critical for traction return and signalling continuity.
47	Curve Rail Lubrication System	Planned	RDSO advisory	Auto-lubricator on curves R<600m. Reduces lateral wear 40-60%.
48	Complete P-Way Renewal (TRT)	Programme	Para 302	Full renewal: rail+sleepers+ballast+realignment. Triggered at GMT limit.

15. AI Assessment Engine

Architecture

The AI Assessment tab uses a **deterministic rule-based engine** that synthesises all computed physics data into a structured P-Way report. No API key or external connection is required. The

engine runs in 400ms (simulated delay for UX) entirely within the browser.

Seven-Section Report Structure

Section 1 — Condition Summary: Concatenates: section name, division, IR zone, climate zone, rail specification, weld type, route class, installation year and age, cumulative GMT and percentage consumed, rail stage, head wear mm and percentage of limit, and Health Index.

Section 2 — Primary Risk: Ranked list of active risks with specific quantitative values and IRPWM paragraph references. Active conditions evaluated:

- Fatigue crack with RUL < 90 days → IRPWM Para 230
- CWR thermal CRITICAL → IRPWM Para 261 (destressing)
- CWR thermal HIGH → IRPWM Para 261 (monitoring)
- GMT > 85% → IRPWM Para 302 (renewal)
- Wear fraction > 75% → renewal
- Hertz shakedown exceeded → ratcheting regime warning
- Ballast < 260mm → IRPWM Para 252
- Maintenance age > 180d → IRPWM Para 235

Section 3 — Immediate Actions (<7 days): Emergency actions triggered by:

- Crack RUL < 30d → "IMPOSE TSR IMMEDIATELY. Notify DE. USFD within 24h." (IRPWM Para 230, 231)
- Crack RUL 30–90d → "USFD scan within 7d. Impose TSR [calculated speed]km/h" (Para 230)
- Thermal CRITICAL → "Destress CWR immediately. TSR 30km/h. Patrol 10:00–17:00 IST."
- Ballast < 240mm → "BCM/BPM deployment within 7d"
- Default → "No emergency action. Continue scheduled cycle."

Section 4 — Maintenance Programme (7–90 days): Condition-triggered long-duration actions:

- Maint > 90d → Tamping with CSM/DTS (IRPWM Para 235)
- GMT > 85% → Urgent renewal work order
- GMT 60–85% → Annual block renewal planning
- Grinding enabled, GMT > 30% → Preventive grinding within 90d
- Curvature > 1.5° → Gauge/cross-level/creep checks + lubrication (Para 218)
- SWR weld type → Joint gap/fish-bolt/joint sleeper check (Para 221)

Section 5 — TSR (Temporary Speed Restriction): Calculated from:

- Crack RUL < 30d → $TSR = \min(20, \text{round}(\text{DAF} \times 15)) \text{ km/h}$

- Thermal CRITICAL → TSR = 30km/h during 10:00–17:00 IST
- GMT > 90% → TSR = round(route_class_max × 0.6) km/h
- Otherwise → "TSR: Not required at present parameters"

Section 6 — Inspection Schedule: Frequency determined by risk level:

- Crack RUL < 60d → Weekly USFD + daily visual patrol
- GMT > 70% → Fortnightly visual + monthly USFD
- Stage ≥ 2 → Monthly visual + quarterly USFD
- Otherwise → Quarterly visual per IRPWM Para 235

Section 7 — Physics Summary: Machine-readable summary of all computed physics values: Hertz p_0 , Winkler σ_b and DAF, Paris Law dK and RUL, CWR σ_{th} and risk level.

16. Dashboard Architecture

Tab Structure

Tab	Contents	Data Source
Track Input	IR Zone/Division selection, section details, 12 track parameters	User input
GMT Tracker	GMT accumulation chart, wear depth chart, train load breakdown, key stats	Physics: ph_gmt(), ph_wear(), buildTL()
Thermal	Annual CWR stress chart, thermal risk tags, seasonal table	Physics: ph_thermal(), ZONES data
Health Analysis	Health gauge, ML inference box, actionable insights, stress utilisation bars, Paris Law trajectory chart	All physics + ML
Defect & Repair	Stage card, governing conditions ranked, top-6 ML-ranked interventions	Stage logic + METHODS
AI Assessment	Generate report button, full text assessment	Rule-based callAI()

Computed Variable Dependencies

```
gmt = ph_gmt(rcKey, ageYrs)
wearR = ph_wear(gmt.cum, curv, rk)
timeline = buildTL(rcKey, rk, yr0, curv)
rail_t = zone.amb[month] + zone.sol[month]
thR = ph_thermal(rail_t, zone.SFT)
hz = ph_hertz(axle)
```

```
wk = ph_winkler(axle, rk, qual, vmax)
dSig = wk.dSig + (thr ? |thr.sig| × 0.30 : 0)
estFreq = fr×86 + pa×96 + em×48
prR = ph_paris(crack, dSig, estFreq, curv)
stageIdx = getStage(gmtF, crackF)
topRepairs = getTopRepairs(stageIdx, crack, thr.risk, curv, ballast, maint,
gmtF)
health = 100 - (d_gmt + d_wear + d_maint + d_ballast + d_crack + d_curv)
```

Technology Stack

Layer	Technology	Version	Purpose
UI Framework	React	18.2.0	Component rendering and state management
Charts	Recharts	2.8.0	Area, Line, Bar, Composed charts
Build Tool	esbuild	0.28.1	JSX compilation + module bundling
ML (Python validation)	scikit-learn	1.8.0	GradientBoostingRegressor/Classifier
Delivery	Single HTML file	—	919KB, zero network dependencies

Station Validation

68 divisions are mapped to 4–6 major stations each. The station dropdown is filtered to the selected division, preventing cross-zone station entries (e.g., cannot select Chennai Central under Delhi Division). This eliminates the most common data entry error in field use — entering a reference station from the wrong division.

17. References

Rail Material and Fracture Mechanics

1. **Indian Railways Standard Specification for Rails (IRS T12-2009).** Research Designs and Standards Organisation (RDSO), Lucknow. Specifies chemical composition, mechanical properties, dimensional tolerances and hardness requirements for GR1080 and other rail grades used on Indian Railways Broad Gauge.
2. **Nippon Steel Corporation (2018). K003en — Rail Products: Track Test Data and Wear Performance.** Nippon Steel Technical Report. Tokyo. Provides field-measured wear rates for DHH370 head-hardened rail under heavy haul and mainline conditions at R=290m and tangent track; used to calibrate wear coefficients (0.013, 0.025, 0.044 mm/MGT).
3. **Paris, P. C., & Erdogan, F. (1963). A Critical Analysis of Crack Propagation Laws.** *Journal of Basic Engineering*, 85(4), 528–533. Foundational paper establishing the power-law relationship $da/dN = C(\Delta K)^m$ for fatigue crack growth. RHIE uses $m=3.0$ as the

standard value for pearlitic rail steel.

4. **Zerbst, U., Mädler, K., & Hintze, H. (2005). Fracture Mechanics in Railway Applications — An Overview.** *Engineering Fracture Mechanics*, 72(2), 163–194. Provides experimentally verified Paris Law constants for rail steel, including $C=1.65 \times 10^{-11}$ m/cycle $\cdot$ (MPa $\sqrt$ m) $^{-3}$ for GR1080-equivalent pearlitic steel under Mode I loading.
5. **Croft, B., Baldry, M., & Christoforou, S. (2016). Fracture Toughness of Rail Steels.** *Proceedings of the 11th International Heavy Haul Association Conference*, Perth. Reports K_{IC} values of 22–35 MPa $\sqrt$ m for pearlitic rail steels at ambient temperature; RHIE adopts the conservative $K_{IC} = 29$ MPa $\sqrt$ m for Mode I transverse cracks.
6. **Agan, H. A., Liu, Z. Y., Hassan, M. F., & Fateh, M. (2004). Mechanical and Fracture Behaviour of Head-Hardened Bainitic Rail Steels.** *Journal of Materials Science*, 39, 6025–6034. Corroborating fracture toughness data for high-strength rail steels.
7. **Johnson, K. L. (1985). Contact Mechanics.** Cambridge University Press. Cambridge. Primary reference for Hertzian contact theory applied to wheel-rail contact; Chapter 7 covers elliptical contact patches and sub-surface shear stress distributions used in the shakedown analysis.
8. **Bower, A. F., & Johnson, K. L. (1989). The Influence of Strain Hardening on Cumulative Plastic Deformation in Rolling and Sliding Line Contact.** *Journal of the Mechanics and Physics of Solids*, 37(4), 471–493. Establishes the elastic shakedown limit criterion used in RHIE ($p_{0_shakedown} \approx 1062$ MPa for GR1080).

Track Mechanics and Structural Analysis

1. **Timoshenko, S. P. (1940). Strength of Materials, Part II.** Van Nostrand. New York. Provides the Winkler elastic foundation beam solution (Chapter 1) used to compute bending moment and rail foot stress under wheel loading: $M = P/(4\lambda)$, $\lambda = (u/4EI)^{0.25}$.
2. **Winkler, E. (1867). Die Lehre von der Elasticitaet und Festigkeit.** Dominicus, Prague. Original derivation of the elastic foundation model (Bettstrahl/beam on elastic foundation) underlying the Winkler track model.
3. **Esvelde, C. (2001). Modern Railway Track (2nd ed.).** MRT Productions. Delft. Comprehensive reference covering CWR thermal stress analysis (Chapter 8), rail fracture mechanics, and track modulus calculations. IRPWM §7 destress thresholds ($\pm 23^\circ\text{C}$, $\pm \Delta T$ limits) are consistent with the criteria detailed in this text.
4. **Grassie, S. L. (2009). Rail Corrugation: Characteristics, Causes and Treatments.** *Proceedings of the Institution of Mechanical Engineers, Part F: Journal of Rail and Rapid Transit*, 223(6), 581–596. Background on corrugation initiation and growth mechanisms informing Stage 1 classification thresholds.

Indian Railways Standards and Codes

1. **Indian Railways Permanent Way Manual (IRPWM), 2021 Edition.** Ministry of Railways, Government of India. The primary maintenance code for all Indian Railways track. Specific paragraphs referenced throughout RHIE:

- **Para 207:** Gauge standards and corrections
 - **Para 208:** Manual packing and emergency spot repairs
 - **Para 214–218:** Ballast specifications, profiling, cleaning, renewal (BG spec: 300mm depth)
 - **Para 220–222, 222A:** Tamping procedures, CSM, DTS
 - **Para 225:** Track slewing and alignment
 - **Para 230–231:** USFD inspection, crack detection, TSR imposition
 - **Para 235–237:** Sleeper fastenings, elastic clips, rail pads
 - **Para 239–246:** Sleeper renewal, anti-creep anchors
 - **Para 248–250:** IRJ renewal, rail bonding
 - **Para 252:** Ballast depth requirements and deficiency remediation
 - **Para 256–269:** Flash butt welding, thermite welding procedures
 - **Para 270–280:** Emergency rail repairs, crack arrest, cropping
 - **Para 302:** GMT life limits for rail renewal (60 kg/m: 800/1000 MGT; 52 kg/m: 525/700 MGT)
 - **Para 318–335:** CWR destressing, SFT, SEJ installation
 - **Para 360–385:** Points and crossings renewal
 - **Para 403–408:** Guard rails, check rails
 - **§7:** CWR distress criteria ($\pm 23^{\circ}\text{C}$ threshold)
2. **Indian Railways Track Machine Manual (IRTMM).** RDSO, Lucknow. Procedures for tamping machines (CSM, DTS), ballast cleaning machines (BCM), and ballast profiling machines (BPM).
 3. **RDSO/2018/Track-BD/01 — Guidelines for Rail Grinding on Indian Railways.** Research Designs and Standards Organisation, Lucknow. Specifies grinding intervals (30–50 MGT on Group A+), metal removal depths for preventive vs corrective grinding, and the GMT life extension achievable (800 → 1000 MGT for 60 kg/m rail).
 4. **RDSO GE-0014/2019 — Guidelines for Geotextile and Geogrid Applications in Railway Track.** Research Designs and Standards Organisation, Lucknow. Referenced for subgrade reinforcement repair method.
 5. **RDSO Corrosion Advisory (2020) — Cathodic Protection for Railway Infrastructure in Coastal and High-Humidity Zones.** Research Designs and Standards Organisation, Lucknow. Referenced for cathodic protection in coastal climate zones (7, 15, 17, 19, 20).

Climatology and Geophysics

1. **India Meteorological Department (2021).** *Climatological Normals 1991–2020*. Ministry of Earth Sciences, Government of India. New Delhi. Primary source for all monthly mean

temperature data used in the 20 climate zone definitions. The 30-year normals (1991–2020) replace the older 1961–1990 normals to capture recent warming trends relevant to CWR management.

2. **Bureau of Indian Standards (2016). IS 1893 (Part 1): 2016 — Criteria for Earthquake Resistant Design of Structures.** BIS, New Delhi. Source for seismic zone classifications (II through V) assigned to each climate zone.
3. **National Disaster Management Authority (2019). Guidelines for Landslide Hazard Zonation Mapping in India.** Government of India. Referenced for terrain risk assessments in Northeast Hill zone (Zone 10) and Western Ghats (Zone 19).

Machine Learning

1. **Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine.** *Annals of Statistics*, 29(5), 1189–1232. Foundational paper for the Gradient Boosting algorithm used in RHIE's ML model. Establishes the stagewise additive modelling framework and residual fitting procedure.
2. **Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python.** *Journal of Machine Learning Research*, 12, 2825–2830. Reference for the scikit-learn 1.8.0 implementation used for Python-side model validation (GradientBoostingRegressor, GradientBoostingClassifier, KFold, cross_val_score).
3. **Brundage, M., et al. (2017). Scalable Learning for Safety-Critical Systems.** Cited as background for the approach of using physics-derived synthetic training data when real labelled inspection records are unavailable — a common practice in safety-critical engineering domains where historical data is proprietary or scarce.

Software and Tools

1. **Meta Platforms / React Contributors (2023). React 18.2.0.** <https://react.dev>. UI component library used for the dashboard frontend. MIT Licence.
2. **Recharts Team (2023). Recharts 2.8.0 — A Composable Charting Library Built on React Components.** <https://recharts.org>. Used for all Area, Line, Bar, and Composed charts in the dashboard. MIT Licence.
3. **esbuild Contributors (2024). esbuild 0.28.1 — An Extremely Fast JavaScript Bundler.** <https://esbuild.github.io>. Used to bundle React, ReactDOM, Recharts, and the RHIE application into a single self-contained HTML file. MIT Licence.
4. **Babel Contributors (2024). @babel/core + @babel/preset-react.** <https://babeljs.io>. Used to transpile JSX syntax to plain JavaScript prior to esbuild bundling.